

Problem

Transform the following sinusoids to phasors:

(a) $-20 \cos(4t + 135^\circ)$

(b) $8 \sin(20t + 30^\circ)$

(c) $20 \cos(2t) + 15 \sin(2t)$

Step-by-step solution

Step 1 of 3

(a)

Consider the following data:

Given function is $-20 \cos(4t + 135^\circ)$

$$\begin{aligned} f &= -20 \cos(4t + 135^\circ) \\ &= -20 \angle 135^\circ \end{aligned}$$

Therefore, the cosine form f is $\boxed{-20 \angle 135^\circ}$.

Step 2 of 3

(b)

Consider the following data:

Given function is $8 \sin(20t + 30^\circ)$

$$\begin{aligned} f &= 8 \sin(20t + 30^\circ) \\ &= 8 \cos(20t + 30^\circ - 90^\circ) \\ &= 8 \angle 30^\circ - 90^\circ \\ &= 8 \angle -60^\circ \end{aligned}$$

Therefore, the cosine form f is $\boxed{8 \angle -60^\circ}$.

Step 3 of 3

(c)

Given function is $20 \cos(2t) + 15 \sin(2t)$

$$\begin{aligned} f &= 20 \cos(2t) + 15 \sin(2t) \\ &= 20 \cos(2t) + 15 \cos(2t - 90^\circ) \\ &= 20 \angle 0^\circ + 15 \angle -90^\circ \\ &= 20 - j15 \end{aligned}$$

$$= 25 \angle -36.86^\circ$$

Therefore, the cosine form f is $\boxed{25 \angle -36.86^\circ}$.